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Inventor(s)	Christian; Paul

TORSION HOLDER

Abstract

A conveyor belt scraper which has a base, a torsion holder with a scraping edge which is mounted to the base for limited rotational movement relative to the base, and a resiliently deformable locking member which restricts said limited rotational movement during use of the belt scraper on a conveyor belt.

Inventors:	Christian; Paul (Booyens, ZA)
Applicant:	Brelko Patents (PTY) Ltd. (Booyens, ZA)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/ZA2024/050001 filed on Jan. 10, 2024, which claims priority to South African Application No. 2023/08925 filed on Sep. 21, 2023, which are hereby incorporated by reference in their entireties.

BACKGROUND

[0002] This invention relates to a conveyor belt scraper.

[0003] US 2018/229943 discloses a conveyor belt scraper wherein a torsion holder with a belt scraping edge is pivotally movable about an axis of a member to which the torsion holder is mounted. In U.S. Pat. No. 6,948,609 discloses a similar feature. Neither citation discloses an easy to implement procedure for aligning the torsion holder with a base to which the holder is fixed.

SUMMARY

[0004] The invention provides a conveyor belt scraper comprising a base which includes a retention formation configured to be engaged with an external support and a first mounting member, a torsion holder, with a belt scraping edge, which is engaged with the first mounting member and which is rotatable to a limited extent relative to the first mounting member, and a locking member which, in use, restricts rotational movement of the torsion holder relative to the first mounting member.

[0005] The first mounting member may include an axis and an outer surface which is radially spaced from the axis. The outer surface may be partly circular cylindrical.

[0006] The torsion holder may include a channel formation with an inner surface which is of complementary form to the circular cylindrical outer surface of the first mounting member.

[0007] A passage may be formed between opposing surfaces of the first mounting member and of the torsion holder e.g. by means of a recess and a hollow in the respective surfaces, and the locking member may be engaged therewith. Preferably the passage is between said inner surface and said outer surface.

[0008] The locking member may be resiliently deformable. The locking member may be elongate circular cylindrical.

[0009] The base may include a bore which extends axially through the first mounting member, and the torsion holder may include a first locating structure with a first locating passage which is axially aligned with the bore.

[0010] A shaft may be located in the bore and in the first locating passage.

[0011] The base may include at least a second mounting member which is spaced axially from the first mounting member. The first locating structure may be positioned between the first mounting member and the second mounting member.

[0012] Preferably the base includes a third mounting member which is spaced axially from the second mounting member and the torsion holder includes a second locating structure with a second locating passage which is axially aligned with the bore and the first locating passage.

[0013] The second mounting member and the third mounting member may be identical to the first mounting member but displaced axially from the first mounting member along the axis. Thus each mounting member may respectively include an outer surface (as described) and a recess in the outer surface.

[0014] Respective bores in the respective mounting members are coaxial and receive the shaft which also extends through the first locating passage and the second locating passage.

[0015] The locking member is operable to deform and allow a limited degree of rotational movement of the torsion holder arm, relative to the base, to take account of forces exerted on the scraping edge by material on a conveyor belt which is being cleaned and, thereafter, as the

deforming force is released to act on the torsion holder arm and thereby restore the arm to a correct belt scraping position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The invention is further described by way of example with reference to the accompanying drawings in which:

[0017] FIG. **1** is a perspective view in exploded form of components of a conveyor belt scraper according to the invention,

[0018] FIG. **2** is a side view on an enlarged scale compared to FIG. **1**, in cross section, of the conveyor belt scraper in an assembled configuration,

[0019] FIG. **3** shows in perspective a lower side of a torsion holder arm included in the conveyor belt scraper; and

[0020] FIG. **4** shows a modified scraper tip for the conveyor belt scraper.

DETAILED DESCRIPTION

[0021] FIG. **1** of the accompanying drawings illustrates in perspective components of a conveyor belt scraper **10** according to the invention, in exploded form.

[0022] The conveyor belt scraper **10** includes a torsion holder **12**, end caps **14**, a base **16**, a locking member **18** and a shaft **20**.

[0023] The locking member **18** is made from a resiliently deformable material such as a suitable grade of rubber, or of polyurethane.

[0024] The torsion holder **12** includes a torsion holder arm **24**, an upper casting **26** which is engaged with the arm and a scraper tip **28** which is fixed to the casting **26**.

[0025] The scraper tip **28** and the casting **26** are made from respective hard-wearing abrasion-resistant materials and comprise a belt scraping edge.

[0026] On a lower inner surface, see FIG. **3**, the torsion holder arm **24** has two locating structures **34** and **36** respectively which are axially spaced apart from each other by a distance **40**. Each locating structure is formed with a respective locating passage **42**. These passages are axially aligned with each other. A downwardly facing hollow **44** which in cross section is semicircular, lies above the locating structures. The semicircular shape is preferred but a hollow with a different cross sectional shape may also function satisfactorily.

[0027] The locating structures **34**, **36** are flanked by wing sections **46A** and **46B** respectively which define a downwardly facing channel formation **46** within which the locating structures **34**, **36** are positioned.

[0028] The end caps **14** are designed to engage with a tight fit with respective open ends **48** of the arm **24**.

[0029] The base **16** has retention formations **50** which are engageable with a sliding action with an external support **52** which is of complementary form to the base, as is shown in FIG. **2**. This arrangement allows the conveyor belt scraper **10** to be correctly positioned across the width of a conveyor belt upon which the belt scraper **10** acts.

[0030] Three mounting members **56**, **58** and **60** respectively are located on upper sides of the retention formations **50**. These mounting members are substantially identical to each other. The first and second members **56** and **58** are spaced apart by a distance **64**. The second and third mounting members **58** and **60** are spaced apart by a distance **68** which is equal to the distance **64**. Each mounting member has a width **70**. The dimension **70** is slightly less than the distance **40** shown in FIG. **3**.

[0031] Each mounting member **56**, **58**, **60** has a respective bore **72** of circular cross section, and a respective recess **74** in an upper part of an outer surface **76**. The bores are in axial alignment with

one another. The recesses **74** are aligned with each other in a longitudinal direction. An upper portion of the outer surface **76** has a generally circular cylindrical shape except for the recess **74**. The outer surface **76** is centred on and is radially spaced from a longitudinally extending axis **80** which passes through the aligned bores **72** in the mounting members.

[0032] Each bore **72** has a diameter which is equal to the diameter of each locating passage **42** and of the shaft **20**.

[0033] When the torsion holder arm **24** is positioned on the base **16** the locating structures **34** and **36** fit closely between the first mounting member **56** and the second mounting member **58**, and between the second mounting member **58** and the third mounting member **60**, respectively. The locating passages **42** in the locating structures are axially aligned with the bores **72** in the three mounting members. The shaft **20** is inserted, moving along the axis **80**, into the aligned bores **72** and the locating passages **42**.

[0034] The wing sections **46A** and **46B** extend downwardly on opposed sides of the mounting members **56**, **58** and **60**, and of the locating structures **34**, **36**.

[0035] The recesses **74**, on upper parts of the outer surfaces **76**, oppose the downwardly facing hollow **44** in the torsion holder arm **24** thereby to form respective passages **84** which are in axial alignment and which extend parallel to the axis **80**. The passages **84** have diameters which are equal to the outer diameter of the locking member **18**. The locking member **18** is then pushed into the passages **84**. The end caps **14** are engaged with the open ends **48** of the torsion holder arm **24** thereby to seal its interior against the ingress of dirt and foreign material.

[0036] FIG. 2 shows that the inner sides of the wing sections **46A** and **46B** and of a cross piece **82**, which is above the mounting members and the locating structures, abut circular portions **76A**, **76B** and **76C** of the outer surfaces **76** of the mounting members and locating structures.

[0037] FIG. 2 illustrates, in an assembled form, the use of the casting **26** to which the scraper tip **28** is fixed. The casting **26** is, in turn, securely attached to an upper end of the torsion holder arm **24**. In an alternative arrangement shown in FIG. 4 the casting **26** is not employed and the scraper tip **28** is located in a slot **28A** which extends the width of the torsion holder arm and the tip is thereby directly secured to the torsion holder arm **24**.

[0038] The channel formation **46** extends around portions of the outer surfaces of the mounting members.

[0039] In use the conveyor belt scraper **10** is mounted so that the scraper tip **28** is in scraping contact with a conveyor belt surface (not shown) which is to be cleaned. This aspect is conventional and therefore is not further described herein.

[0040] In use of the FIG. 2 embodiment or the FIG. 4 embodiment, force which is exerted on the scraper tip **28**, and hence on the arm **24** by movement of the conveyor belt, tends to rotate the torsion holder arm **24** in a clockwise direction about the axis **80** (referring to FIG. 2). The locking member **18**, which is made from a resiliently deformable material such as a hard rubber or polyurethane, acts to resist such rotation but can deflect slightly as the force on the scraper tip increases. The locking member **18** then restores the arm **24** to a correct scraping position with the tip **28** in scraping contact with the conveyor belt surface. This process allows an effective belt scraping action to be maintained continuously but with the capability of the scraping edge to deflect slightly when the force which is exerted on the scraping edge by material on the conveyor belt increases.

[0041] The inner sides of the wing sections **46A** and **46B** rotate slightly about the circular outer surfaces **76** of the mounting members **56**, **58** and **60**. As noted the locking member **18** deforms slightly with this movement. This rotating movement is accompanied by a corresponding small degree of rotation of the locating structures **34**, **36** which are between the associated pairs of the mounting members, about the shaft **20** which is centred on the axis **80**.

[0042] The extent of this relative rotational movement is restrained or controlled by the size and by the type of material of the locking member **18** which is deformed slightly as the torsion holder arm

tends to rotate about the mounting members on the base and which then restores the torsion holder arm to a correct scraping position.

[0043] In cross section the locking member **18** is of annular form and its deformation characteristics are additionally dependent on the size of a central passage **18A** and on the thickness of a wall **18B** around the passage **18A**. The external diameter of the locking member, the thickness of the wall **18B**, the diameter of the passage **18A** and the material from which the locking member is made are important parameters which are chosen to control the force versus deflection relationship, in a circular sense, of the torsion holder arm.

[0044] The conveyor belt scraper **10** in use thus, to the extent possible, is held in firm scraping contact with a moving conveyor belt surface. Although the permitted degree of deflection of the belt scraping edge of the scraper tip is limited by the locking member **18**, when a hard material on the belt surface is brought into contact with the belt scraping edge, damage to the scraper **10** is generally restricted without adversely affecting scraping efficiency.

Claims

1. A conveyor belt scraper comprising a base which includes a retention formation configured to be engaged with an external support and at least a first mounting member, a torsion holder, with a belt scraping edge, which is engaged with the first mounting member and which is rotatable to a limited extent relative to the first mounting member, and a locking member which, in use, restricts rotational movement of the torsion holder relative to the first mounting member wherein the base includes a bore which extends axially through the first mounting member, and the torsion holder includes a first locating structure with a first locating passage which is axially aligned with the bore, and a shaft which is located in the bore and in the first locating passage.
 2. The conveyor belt scraper of claim 1 wherein the first mounting member includes an axis and an outer surface which is partly circular cylindrical.
 3. The conveyor belt scraper of claim 2 wherein the torsion holder includes a channel formation with an inner surface which is of complementary form to the circular cylindrical outer surface of the first mounting member.
 4. The conveyor belt scraper of claim 1, wherein a passage is formed between opposing surfaces of the first mounting member and of the torsion holder and the locking member is engaged with the passage.
 5. The conveyor belt scraper of claim 4, wherein the locking member is resiliently deformable.
 6. The conveyor belt scraper of claim 5, wherein the locking member is elongate circular cylindrical and in cross section is of annular form.
 7. The conveyor belt scraper of claim 1, wherein the base includes a second mounting member which is spaced axially from the first mounting member, the bore extending axially through the second mounting member, and the first locating structure is positioned between the first mounting member and the second mounting member.
 8. The conveyor belt scraper of claim 7 wherein the base includes a third mounting member which is spaced axially from the second mounting member, the bore extending axially through the third mounting member and the torsion holder includes a second locating structure which is positioned between the second mounting member and the third mounting member, the second locating structure including a second locating passage which is axially aligned with the bore and the first locating passage and the shaft is located in the first locating passage and the second locating passage and in the respective bores in the respective mounting members.
 9. The conveyor belt scraper of claim 8, wherein the locking member is resiliently deformable and restricts rotational movement of the torsion holder relative to the mounting members.
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